LIFTING AN IDEMPOTENT MODULO AN IDEAL AFTER AN ETALE BASE CHANGE THAT IS AN ISOMORPHISM ON THE QUOTIENT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

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Allowed standard prerequisites (editorial).

- Existence of a monic polynomial modulo the ideal vanishing at a lift of the idempotent. Lemma \ref{lemma-helper-integral}, cited at proof lines 1988-1989. Produces the polynomial f whose factorisation drives the construction.
- Factorisation of that polynomial after an etale base change inducing an isomorphism modulo the ideal. Lemma \ref{lemma-lift-factorization-easy}, cited at proof lines 1990-1992. Gives the two factors g and h congruent to a variable power and to a power of the variable minus one modulo the ideal.
- An integral ring map is closed on spectra, and closed disjoint loci can be separated by an element. Algebra, Lemmas \ref{algebra-lemma-integral-going-up} and \ref{algebra-lemma-going-up-closed}, cited at proof lines 2007-2009, and Lemma \ref{lemma-separate-image-closed-from-closed}, cited at proof lines 2010-2012. Produce the element whose localisation makes the two loci cover the spectrum.
- Idempotents correspond to open and closed subsets, and the associated idempotent is unique. Algebra, Lemma \ref{algebra-lemma-disjoint-decomposition}, cited at proof lines 2014 and 2017. Give the idempotent of the disjoint decomposition and identify it as the required lift.

Source reference. Original lemma 07M4: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9714–9714. The lemma's own canonical tag, quoted

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so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 09XG: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 12529–12529. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 07M1: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9711–9711. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 00GU: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 620–620. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 00HZ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 661–661. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 07M3: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9713–9713. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 00EE: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 531–531. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let A be a ring, let $I \subset A$ be an ideal. Let $A \to B$ be an integral ring map. Let $\overline{e} \in B/IB$ be an idempotent. Then there exists an étale ring map $A \to A'$ which induces an isomorphism $A/I \to A'/IA'$ and an idempotent $e' \in B \otimes_A A'$ lifting $\overline{e}$.*

Proof. Choose an element $y \in B$ lifting $\overline{e}$. Choose $f \in A[x]$ as in Lemma 09XG for y. By Lemma 07M1 we can find an étale ring map $A \to A'$ which induces an isomorphism $A/I \to A'/IA'$ and such that $f = gh$ in $A[x]$ with $g(x) = x^d \bmod IA'$ and $h(x) = (x-1)^d \bmod IA'$. After replacing A by A' we may assume that the factorization is defined over A. In that case we see that $b_1 = g(y) \in B$ is a lift of $\overline{e}^d = \overline{e}$ and $b_2 = h(y) \in B$ is a lift of $(\overline{e} - 1)^d = (-1)^d(1 - \overline{e})^d = (-1)^d(1 - \overline{e})$ and moreover $b_1 b_2 = 0$. Thus $(b_1, b_2)B/IB = B/IB$ and $V(b_1, b_2) \subset \mathrm{Spec}(B)$ is disjoint from $V(IB)$. Since $\mathrm{Spec}(B) \to \mathrm{Spec}(A)$ is closed (see Algebra, Lemmas 00GU and 00HZ) we can find an $a \in A$ which maps to an invertible element of A/I whose image in B lies in (b_1, b_2), see Lemma 07M3. After replacing A by the localization A_a we get that $(b_1, b_2) = B$. Then $\mathrm{Spec}(B) = D(b_1) \amalg D(b_2)$; disjoint union because $b_1 b_2 = 0$ and covers $\mathrm{Spec}(B)$ because $(b_1, b_2) = B$. Let $e \in B$ be the idempotent corresponding to the open and closed subset $D(b_1)$, see Algebra, Lemma 00EE. Since b_1 is a lift of $\overline{e}$ and b_2 is a lift of $\pm(1 - \overline{e})$ we conclude that e is a lift of $\overline{e}$ by the uniqueness statement in Algebra, Lemma 00EE. □